Exercise 3.8 (*)

Let $h : \mathbb{R} \to \mathbb{R}$ be the 1-periodic function satisfying $h(x) = |x|$ for $x \in [-\frac{1}{2}, \frac{1}{2}]$ (and $h(x+1) = h(x)$). Define the Takagi function

$$T(x) = \sum_{i=0}^{\infty} \frac{1}{2^i} h(2^i x).$$

(a) *Sketch h.* On $[-\frac{1}{2}, \frac{1}{2}]$ it is the V–shape $|x|$ (cusp at 0), extended periodically with period 1. Equivalently,

$$h(x) = \operatorname{dist}(x, \mathbb{Z}) = \min_{k \in \mathbb{Z}} |x - k|,$$

so $0 \le h(x) \le \frac{1}{2}$, attaining 0 at integers and $\frac{1}{2}$ at half–integers.

(b) *Uniform continuity and differentiability.* Since $x \mapsto \operatorname{dist}(x, \mathbb{Z})$ is 1–Lipschitz, h is uniformly continuous on $\mathbb{R}$. On each interval $(k, k + \frac{1}{2})$ and $(k - \frac{1}{2}, k)$ the function is affine with slope -1 or $+1$, hence differentiable there. It fails to be differentiable precisely at points congruent to 0 or $\frac{1}{2}$ modulo 1, i.e. at those x with $2x \in \mathbb{Z}$. Thus h is differentiable at every x with $2x \notin \mathbb{Z}$.

(c) *Uniform convergence of the series and uniform continuity of T.* Since $0 \le h \le \frac{1}{2}$,

$$\left| \frac{1}{2^i} h(2^i x) \right| \le \frac{1}{2^{i+1}} \qquad (\forall x \in \mathbb{R}),$$

and $\sum_{i=0}^{\infty} 2^{-(i+1)} < \infty$. By the Weierstrass M–test, the series converges uniformly on $\mathbb{R}$ to a continuous function T. Each summand $x \mapsto 2^{-i} h(2^i x)$ is uniformly continuous (indeed 1–Lipschitz), hence T, being a uniform limit of uniformly continuous functions, is uniformly continuous on $\mathbb{R}$.

(d) Let $x \in \mathbb{R}$. There exist unique u_n, v_n such that

(i) $2^n u_n,\ 2^n v_n \in \mathbb{Z}$,

(ii) $v_n - u_n = 2^{-n}$,

(iii) $u_n \le x < v_n$.

Solution. Take $u_n = \frac{\lfloor 2^n x \rfloor}{2^n}$ and $v_n = u_n + 2^{-n}$. Then (i)–(iii) hold. Uniqueness: if $u_n' \le x < v_n'$ with (i)–(ii), then $2^n u_n' \le 2^n x < 2^n v_n'$ and $2^n v_n' - 2^n u_n' = 1$, so $2^n u_n' = \lfloor 2^n x \rfloor = 2^n u_n$, hence $u_n' = u_n$ and $v_n' = v_n$.

(e) For $i \ge 0$ set $h_i(x) = 2^{-i} h(2^i x)$. Show that:

(i) $h_i(u_n) = h_i(v_n) = 0$ for $i \ge n$.

(ii) $\dfrac{h_i(v_n) - h_i(u_n)}{v_n - u_n} = \pm 1$ for $i < n$.

Solution. (i) Since $2^n u_n, 2^n v_n \in \mathbb{Z}$, for $i \ge n$ also $2^i u_n, 2^i v_n \in \mathbb{Z}$, hence $h(2^i u_n) = h(2^i v_n) = 0$.

(ii) The map $x \mapsto 2^i x$ sends $[u_n, v_n]$ onto the interval $\left[\frac{K}{2^{n-i}}, \frac{K+1}{2^{n-i}}\right]$ where $K = \lfloor 2^n x \rfloor$. Its length is $2^i(v_n - u_n) = 2^{-(n-i)} \le \frac{1}{2}$, so it contains no half–integer. Hence h is affine with slope ± 1 on this interval, and therefore

$$\frac{h_i(v_n) - h_i(u_n)}{v_n - u_n} = \frac{2^{-i}\big(h(2^i v_n) - h(2^i u_n)\big)}{2^{-n}} = 2^{n-i} \cdot 2^{-i} \cdot (\pm 2^{-(n-i)}) = \pm 1.$$

(f) *Conclude that*

$$\frac{T(v_n) - T(u_n)}{v_n - u_n} = \sum_{i=0}^{n-1} \frac{h_i(v_n) - h_i(u_n)}{v_n - u_n},$$

and that the sum on the right does not converge as $n \to \infty$.

Solution. Since $h_i(u_n) = h_i(v_n) = 0$ for $i \geq n$,

$$T(v_n) - T(u_n) = \sum_{i=0}^{\infty} \big(h_i(v_n) - h_i(u_n)\big) = \sum_{i=0}^{n-1} \big(h_i(v_n) - h_i(u_n)\big).$$

Divide by $v_n - u_n$ to get the displayed identity. By (e)(ii), each term of the sum equals ± 1, so the sequence of partial sums cannot converge (successive differences are $\pm 1 \not\to 0$). Hence the right–hand side does not converge as $n \to \infty$. In particular, T is nowhere differentiable.